Methodology Review: ARIMA and ARIMAX for Forecasting UK Vacancies

1. ARIMA in plain terms

ARIMA forecasts a series using only its own history. Write y_t for vacancies in month t. ARIMA has three ingredients, each answering a simple question; (p, d, q) says how much of each to use.

- **AR ("autoregressive"), order p: do recent values predict the next one?** The forecast is a weighted sum of the last p values. For $p = 1$:

  $$y_t = c + \phi\, y_{t-1} + \varepsilon_t.$$

  Here c is a constant, ϕ says how strongly this month carries over from last month (e.g. $\phi = 0.8$ means 80% of last month's deviation persists), and ε_t is new, unpredictable noise.
- **I ("integrated"), order d: should we model changes rather than levels?** If the series drifts (a trend, or a level that keeps moving), we model the month-to-month change $y_t - y_{t-1}$ instead ($d = 1$). Forecasts are then rebuilt by adding the predicted changes back onto the last value.
- **MA ("moving average"), order q: do recent forecast errors predict the next value?** If the model under-predicted last month, part of that miss often carries into this month. For $q = 1$:

  $$y_t = c + \varepsilon_t + \theta\, \varepsilon_{t-1}.$$

  The weight θ says how much of last month's error is still felt.

Putting it together. ARIMA(p, d, q) first applies differencing d times, then models the result with p AR terms and q MA terms. For example, ARIMA(1,1,1) says:

$$\underbrace{(y_t - y_{t-1})}_{\text{this month's change}} = c + \phi\ \underbrace{(y_{t-1} - y_{t-2})}_{\text{last month's change}} + \theta\ \underbrace{\varepsilon_{t-1}}_{\text{last month's error}} + \varepsilon_t.$$

In words: *this month's change = a constant + a share of last month's change + a share of last month's miss + new noise.* The parameters c, ϕ, θ are estimated from the data. As a special case, ARIMA(0,1,0) is the naive forecast (next value = last value).

Why stationarity matters. A series is stationary if its average level, spread and month-to-month patterns stay the same over time. AR and MA terms learn fixed patterns, which only makes sense if the patterns do not change. If the series trends, the estimated patterns are unreliable and forecasts can be badly off. Differencing is the fix, but it throws away information about the level, and differencing too much adds noise, so use the smallest d that makes the series look stable. If the data are monthly and seasonal, add seasonal terms (SARIMA).

2. ARIMAX: adding the Indeed index

ARIMAX adds an outside variable x_t (the Indeed UK Job Postings Index). The cleanest version (and what Hyndman & Athanasopoulos call a *dynamic regression*) is

$$y_t = \beta_0 + \beta_1 x_{t-k} + \eta_t, \qquad \eta_t \sim \text{ARIMA}(p, d, q),$$

i.e. a regression on x whose leftover part is allowed to be autocorrelated. If $d > 0$, *both* y and x are differenced the same way. Lags $(x_{t-1}, x_{t-2}, \dots)$ let the index act with a delay. Note that `statsmodels` `SARIMAX` fits exactly this "regression with ARIMA errors" form, not a model with lagged y multiplied by x.

Why Indeed might help. Postings are an early signal: employers advertise before the official vacancy survey picks it up, and Indeed data is available quickly while ONS figures arrive weeks later.

Key rule: no look-ahead. A forecast made at time T may only use Indeed data available at T. Two consequences:

- To forecast h months ahead, use only lags $k \geq h$ of x, or else forecast x too (which adds error).
- Where possible, use the data as first published. Revised series flatter the model because they contain information that did not exist at the time.

Assumptions for ARIMAX: (i) the relationship is roughly stable over time; (ii) x drives vacancies, not the reverse (otherwise β is biased and ARIMAX is the wrong tool); (iii) after differencing both series are stationary; (iv) residuals are uncorrelated, with roughly constant variance.

3. Choosing the ARIMA orders

- **Choose d:** plot the series; run a unit-root test (KPSS, ADF) and treat the outcome as a guide, not a verdict. A slowly decaying ACF suggests differencing is needed.

- **ACF / PACF** of the differenced series: the ACF shows correlation between y_t and y_{t-k}; the PACF shows it after removing the in-between lags. A PACF that cuts off after lag p (with a decaying ACF) suggests AR(p); an ACF that cuts off after lag q suggests MA(q). These are only starting points.
- **AIC / BIC:** fit several candidates and compare. Both reward fit and penalise extra parameters; BIC penalises more, so it prefers simpler models, which suits a short sample. Compare only models with the same d and the same data. Use AICc for small samples.
- **Residual checks:** residuals should look like random noise. Plot them over time, check their ACF, run a Ljung–Box test (no leftover autocorrelation) and look at a histogram or Q–Q plot. Leftover patterns mean the model missed structure. For ARIMAX, also check the cross-correlation between residuals and x.

4. Assumptions and difficulties for UK vacancies

- **Stationarity:** vacancies trend and shift with the economic cycle, so expect $d = 1$. Test, do not assume.
- **Seasonality:** check whether the ONS series is already seasonally adjusted, and whether the Indeed index is. Mixing adjusted and unadjusted series creates false relationships.
- **COVID and other shocks:** the 2020 collapse and 2021–22 rebound are extreme outliers that can dominate estimation. Options: dummy variables, excluding the period, or reporting results with and without it.
- **Do the two series measure the same thing?** Indeed covers only one site's postings, which can be duplicated, left open after filling, or skewed toward certain sectors. ONS counts vacancies from an employer survey. Their link may be loose.
- **Short sample:** the Indeed index appears to start in early 2020 (to be confirmed), giving well under a hundred monthly points, nearly all of them post-COVID. Keep models small (few parameters, few lags); complex models will overfit.
- **Indeed need not help:** extra predictors add estimation noise. ARIMAX can forecast worse than plain ARIMA, and the comparison should be allowed to show this.
- **Structural change:** the Indeed–vacancy relationship may shift (e.g. post-pandemic changes in how firms advertise). A single fixed β would then be wrong. This motivates the Term 2 changepoint work; for now, check whether rolling estimates of β are stable.

5. Recommended approach

Baseline. Use the naive forecast (next value = last observed value), with the drift or seasonal-naive variant if the data justify it. Any model that cannot beat this is not useful.

ARIMA. Settle on d via plots and tests; shortlist orders from the ACF/PACF; compare with AICc/BIC and automatic search (Hyndman–Khandakar, e.g. `pmdarima`); keep the simplest model that passes residual checks. Re-estimate at each forecast origin.

ARIMAX. Start from the chosen ARIMA, add the Indeed index at lags $k \geq h$, and pick the lag(s) by AICc/BIC and out-of-sample performance (not by in-sample correlation alone). Keep at most one or two lags given the sample size.

Evaluation. Use *rolling-origin* (expanding-window) forecasts: refit at each origin using only the data available then (including Indeed data as available then), forecast $h = 1, 3, 6$ months ahead, and score all three models on identical origins. Report MAE, RMSE and MASE (error relative to the naive model). Test differences with Diebold–Mariano, bearing in mind its weak power on short samples.

Proposed sequence.

1. Build the naive benchmark and the rolling-origin framework.
2. Examine stationarity and seasonality; select candidate ARIMA orders.
3. Fit the ARIMA baseline and run residual diagnostics.
4. Add lagged Indeed index (no look-ahead); re-run diagnostics.
5. Compare out-of-sample accuracy across naive, ARIMA and ARIMAX, with and without the COVID period.

Points to discuss. (a) Exact start date and revision policy of the Indeed series; (b) whether the ONS series used is the 3-month rolling average, which is smooth and strongly autocorrelated by construction; (c) forecast horizons of interest; (d) treatment of the COVID period.